

SRC Base Operating Instructions

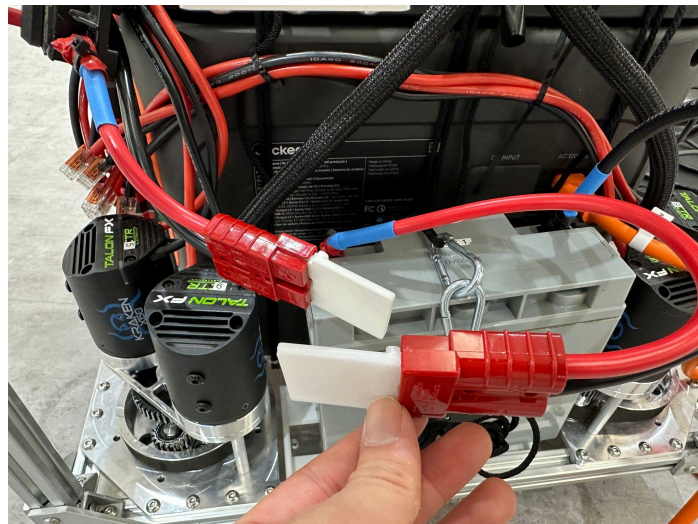
The SRC base consists of the Franka Panda FR3 robot (named Juliet) and the Tidybot++ base. One mini-pc controls both arm and mobile base, where OpenSai controls the arm through torque commands (like in class), and sends desired base pose commands to the base (all through redis).

The Franka arm is on 172.16.0.12.

The computer username is tidybot01, and the password is tidy. To control the mini-pc, a wireless HDMI is paired, with a wireless keyboard setup.

Start Up:

Connect the motor battery supply together. Remember to not lose the plastic inserts. Then, use the remote to trigger the wireless relay to power on the motor power supply.



Then, click on the remote control to enable the relay for the power supply. ALWAYS have the remote control with you, as it is the e-stop for the motors.

The lights for each motor should start blinking.

Turn on the camping battery pack by pressing the two buttons on the front (shown in the image below, where the green LEDs are glowing). This powers the mini-pc and the Franka arm. It takes a few seconds for the camping battery to initialize.



Turn on the mini-pc and the Franka control box (following the instructions for the Franka arm).

Operation:

To run the mobile base, open a terminal and cd into the tidybot2 repo. Then, run “mamba activate tidybot2” in the terminal to activate the virtual environment. From there, there are 2 scripts that are used.

python gamepad_teleop.py is the command to run for the logitech gamepad teleoperation. This should be used for conveniently moving the base around.

python3 redis_driver.py is the command to run for the redis driver. This should be used for your project.

With the redis driver, there are the following keys:

hb1::kill

- Initially set to the string “ok” - set the string to “kill” to stop the driver. This will decelerate the mobile base and terminate the base driver. Note that the global reference frame will be reset every time the driver is killed and restarted.

hb1::stop

- Initially set to the string “ok” - set the string to “stop” to stop the robot. This will decelerate the mobile base and override the desired pose to the current pose. This is used to stop the robot, but not kill the driver.

hb1::desired_pose

- Initially set to the starting pose “[0, 0, 0]” where the command is [x (m), y (m), theta (rad)] in global coordinates. The global reference frame is set to the initial base pose when the driver is run.

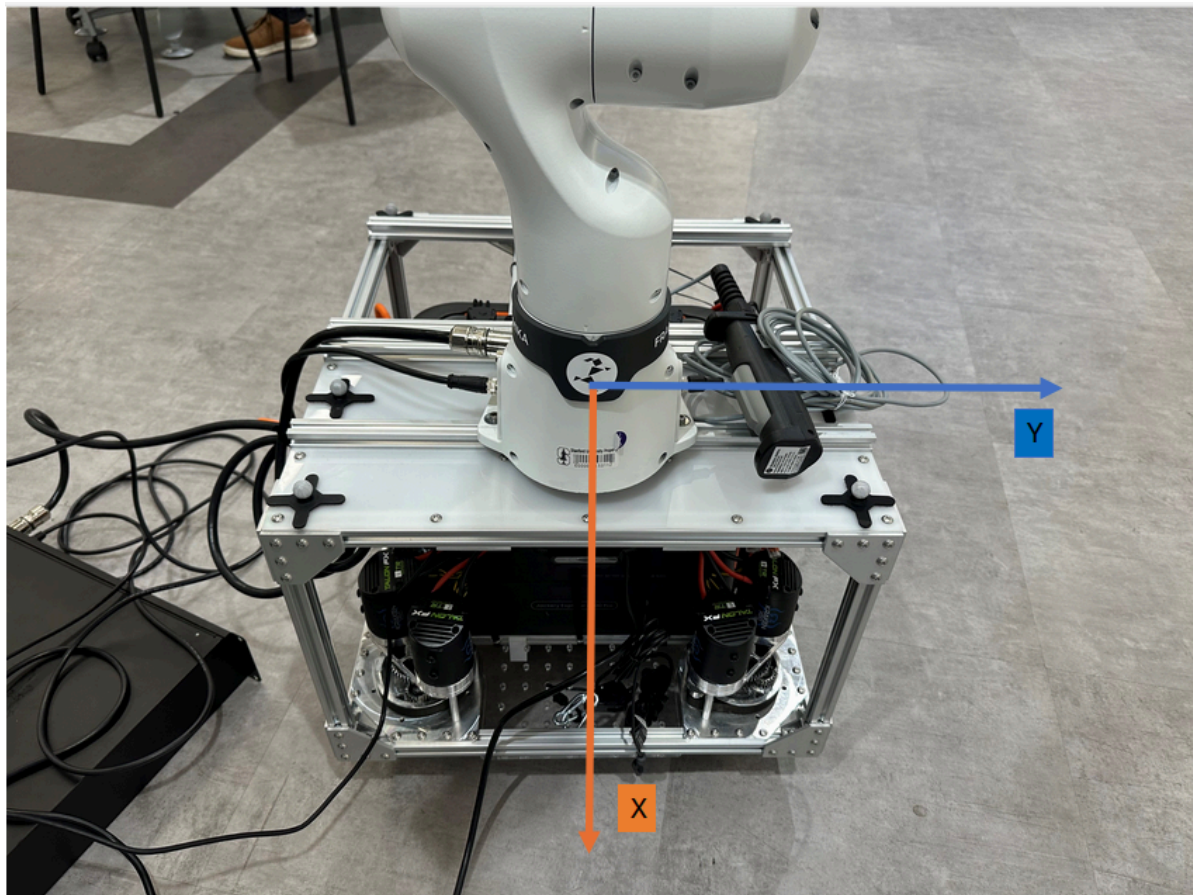
hb1::current_vel

- Provides the current base velocity.

hb1::current_pose

- Provides the current base pose.

The following image shows the coordinate frame of the mobile base, where the initial global reference frame is centered on the mobile base. The Franka arm +X direction is aligned with the mobile base local +X direction.



Shut Down:

To shut down the system properly:

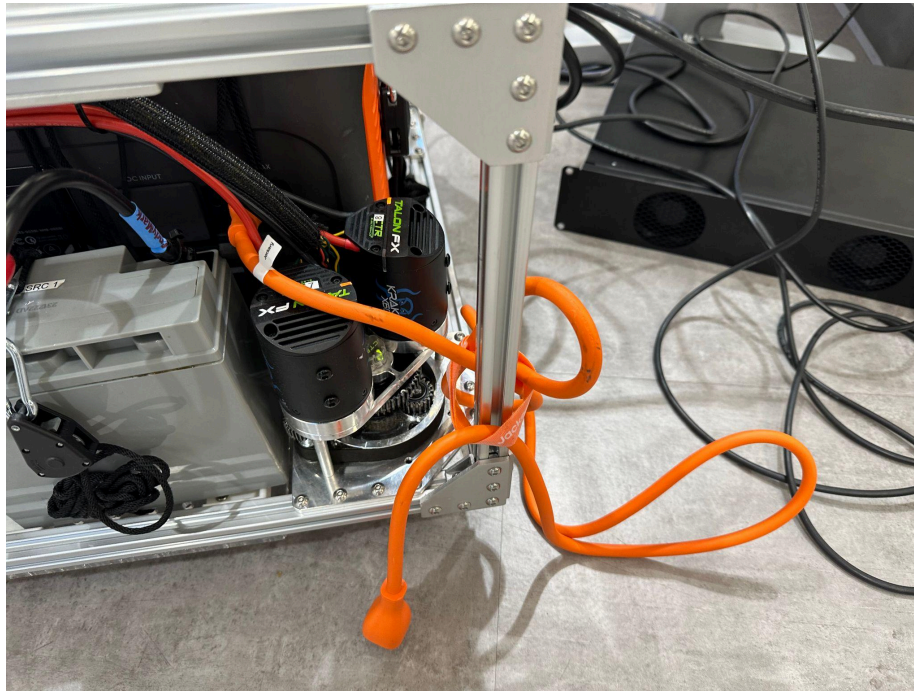
1. Shut down the Franka arm through the GUI interface at 172.16.0.12 (in a web browser)
2. Wait for the shut down message to show, and then turn off the controller box
3. Kill the mobile base driver
4. Turn off the motor power supply via the wireless remote
5. Unplug the motor battery supply, and re-insert the plastic inserts if not charging

Put the mobile base back near the work station, and make sure to charge the camping battery and the motor battery.

Recharging:

The gray batteries to power the motors and the camping battery need to be recharged. To recharge the gray batteries, connect the battery charger in the SRC tool bay to the batteries. To recharge the camping battery, plug the orange power cable to an outlet.





Notes:

- Make sure to check the remaining battery life on the camping battery.
- Make sure to not interfere or disconnect the control box while the robot is powered. This will potentially brick the system.
- Read the output of the redis driver terminal. It will contain error messages.